# Exercise 3(b) – Data Annotation and Feature Engineering

Machine Learning | Group 40

**Question:** Look back to Exercise 2. Which symbol in the picture corresponds to the role that you are playing? Which symbol corresponds to the functions that you implement in this exercise?

## Answer

**1. The role we are playing corresponds to $\gamma(o)$.**

As annotators, we read each text and assign its correct class label: Human or LLM. In the diagram from Exercise 2, $\gamma(o)$ represents the mapping from a real-world object to its class. Therefore, it corresponds to the annotation task.

**2. The functions we implement correspond to $\alpha(o)$.**

In this exercise, we implement feature-extraction functions that convert each raw text into numerical features, such as the number of sentences and average word length. In the diagram, $\alpha(o)$ represents the mapping from a real-world object to its representation in the feature space X. Therefore, it corresponds to the feature-extraction functions.

| Symbol | Meaning in Exercise 3 |
|---|---|
| $\gamma(o)$ | Assigns the correct class label (Human or LLM) to a text. |
| $\alpha(o)$ | Extracts numerical features from a text and maps it into feature space X. |

**Conclusion:** $\gamma(o)$ represents the annotation role because it assigns class labels, while $\alpha(o)$ represents the feature-extraction functions because they transform texts into numerical feature representations.